Response to Reviewers — Second Cut

Team 0 · *A Load-Cell Cup Detection Module*

Acknowledgements

The authors thank the reviewers for their careful evaluation. Line, section, and figure numbers refer to the revised manuscript. **Main changes:** the Introduction now states a research question (Section **??**), and Section **??** adds the module architecture (Fig. **??**).

Summary of changes (observation–response–change matrix)

7 observations: 5 addressed, 1 partially addressed, 1 not addressed (justified).

Key	Status	Location in the revised paper
R1.1	**[Addressed]**	Section **??**, line **??**, line **??**
R1.2	**[Addressed]**	Whole manuscript
R2.1	**[Addressed]**	Lines **??–??**
R3.1	**[Addressed]**	Fig. **??**, Eq. (**??**), lines **??–??**
R3.2	**[Partially addressed]**	Table **??**, lines **??–??**
R4.1	**[Addressed]**	Lines **??–??**
R4.2	**[Not addressed (justified)]**	Lines **??–??**

Reviewer 1 — *Format, LaTeX and template compliance*

R1.1 [Addressed]

Reviewer comment: Las fuentes están escritas a mano en formato autor–año.

Response: We created one BibTeX entry per source and replaced every hand-written citation with `\cite{key}` (Section **??**, line **??**, line **??**).

Changes in the revised paper: Section **??**, line **??**, line **??**

R1.2 [Addressed]

Reviewer comment: Los párrafos terminan con \\.

Response: All forced line breaks were removed; paragraphs are separated by a blank line.

Changes in the revised paper: Whole manuscript

Reviewer 2 — *Writing and structure*

R2.1 [Addressed]

Reviewer comment: El artículo no formula una pregunta de investigación ni declara sus contribuciones.

Response: We added a research question and three contributions at the end of the Introduction (lines **??–??**). The research question now reads:

> *"Can a single load cell under the cup holder detect cup placement and removal reliably enough, and fast enough, to release the robot automatically?"*

Changes in the revised paper: Lines **??–??**

Reviewer 3 — *Technical content*

R3.1 [Addressed]

Reviewer comment: No se presenta la arquitectura del módulo ni la regla de detección.

Response: The architecture is shown in Fig. **??** (lines **??–??**), and the detection rule with two thresholds is given in (**??**) (lines **??–??**).

Changes in the revised paper: Fig. **??**, Eq. (**??**), lines **??–??**

R3.2 [Partially addressed]

Reviewer comment: No hay pruebas que respalden la confiabilidad de la detección.

Response: The test protocol was added in Table **??**, and the first 20 cycles are reported in lines **??–??**. The complete campaign of 100 cycles will be reported in the next cut.

Changes in the revised paper: Table **??**, lines **??–??**

Reviewer 4 — *D-Lab and state of the art*

R4.1 [Addressed]

Reviewer comment: No se explica por qué RFID no es adecuado para RoboCoffee.

Response: Related Work now explains that RFID needs a tag on every disposable cup (lines **??–??**).

Changes in the revised paper: Lines **??–??**

R4.2 [Not addressed (justified)]

Reviewer comment: Implementen una segunda versión del módulo con cámara y comparen ambas.

Response: We agree that a camera could identify the cup independently of its weight. However, image processing and controlled lighting are outside the scope of this module. Instead, vision-based detection is discussed as future work in lines **??–??**.

Changes in the revised paper: Lines **??–??**